1. Brainstorming & Ideation Phase

FitBuddy – AI Fitness Plan Generator

- Problem / Idea: Users need simple, personalized workout guidance aligned with their fitness goals.
- Proposed idea: A web application that uses Generative AI to create customized 7-day workout plans and nutrition/recovery tips.
- Target goals: Weight loss, muscle gain, general wellness and other fitness objectives.
- Core scenarios: Generate a plan from user details; refine the plan through feedback; provide goal-based nutrition/recovery tips.
- Admin scenario: Coaches/administrators can view registered users and their original/updated plans.

2. Requirement Analysis Phase

Requirements & Technology

- User inputs: Name/User ID, age, weight, fitness goal and preferred intensity (Low/Medium/High).
- Functional requirements: Workout generation, nutrition/recovery tip generation, feedback-based plan updating and admin user/plan viewing.
- Frontend: HTML, CSS and Jinja2 templates.
- Backend: FastAPI with Python and Uvicorn.
- AI layer: Google Gemini models — Gemini 1.5 Pro for workout generation/updating and Gemini Flash for fast nutrition tips.
- Database: SQLite with SQLAlchemy ORM for user and plan persistence.

3. Project Design Phase

Solution Design & Architecture

- Frontend layer: index.html collects user data; result.html displays the generated plan, nutrition tip and feedback form; all_users.html provides the admin view.
- Backend layer: FastAPI routes receive form data, validate it and connect the application components.
- AI layer: Gemini Pro generates structured 7-day plans and revises plans from feedback; Gemini Flash produces concise nutrition/recovery tips.
- Data layer: SQLAlchemy + SQLite stores user information and preserves original and updated plans.
- Main flow: User Input → FastAPI → Gemini → Database → Jinja2 Result.

4. Project Planning Phase

Milestones, Activities & Execution Plan

- Milestone 1 – Model Selection & Architecture: evaluate AI models, define architecture and prepare the environment.
- Milestone 2 – Core Functionality: develop workout generation, nutrition tips, feedback updating and storage.
- Milestone 3 – routes.py Development: implement home, generate-workout, submit-feedback and view-all-users routes.
- Milestone 4 – Frontend Development: build responsive HTML/CSS interfaces and dynamic Jinja2 templates.
- Milestone 5 – Deployment: prepare local deployment, install dependencies and verify the application with Uvicorn.
- Development prerequisites: Python, FastAPI, Gemini API familiarity, HTML/CSS/Jinja2, Git, SQLAlchemy/SQLite and virtual-environment setup.

5. Project Development Phase

Implementation & Working Features

- Workout generation: generate_workout_gemini() sends the user's goal/intensity to Gemini 1.5 Pro and returns a structured day-wise plan.
- Nutrition tip: generate_nutrition_tip_with_flash() creates a concise goal-aligned nutrition or recovery suggestion.
- Feedback update: update_workout_plan() combines the original plan with user feedback to create a revised plan.
- Database functions: save_user(), save_plan(), update_plan(), get_original_plan() and get_user() support persistence.
- Routes connect the frontend, Gemini AI functions and database; generated results are rendered through Jinja2.
- UI features include user input, personalized results, feedback submission and an administrator view.

6. Project Testing Phase

Testing, Verification & Expected Results

- Environment verification: create/activate the virtual environment and install dependencies from requirements.txt.
- API/server verification: run the FastAPI application with Uvicorn and check the local application and /docs interface.
- Input testing: verify that user details and intensity values are accepted and processed correctly.
- AI-output testing: verify structured 7-day workout plans, nutrition/recovery tips and feedback-based revisions.
- Database testing: verify that user details, original plans and updated plans are stored and retrieved correctly.
- UI testing: verify navigation, form submission, result rendering and responsive presentation.

7. Project Documentation Phase

Documentation & Deliverables

- Document the project overview, scenarios, architecture, prerequisites, milestones, implementation and deployment steps.
- Document important modules/functions: Gemini generators, routes.py, database.py and Jinja2 templates.
- Document the user workflow: enter details → generate plan → view nutrition tip → submit feedback → receive updated plan.
- Include screenshots/evidence of the homepage, personalized plan, nutrition tip, feedback result and admin users page.
- Maintain project files and documentation in the GitHub repository using the required phase-wise folder structure.
- Submission must include a public GitHub repository link containing the project files/documents.

8. Project Demonstration Phase

Demo, Submission & Future Plan

- Demo video: record the complete project demonstration with screen sharing and student voice-over/explanation.
- Explain in the video: Project Name, Purpose, Uses/Benefits, Execution/Working Process and Final Output.
- Upload the demo video to Google Drive and set access to “Anyone with the link can view”.
- Repository structure: 8 phase folders — Brainstorming & Ideation, Requirement Analysis, Project Design, Project Planning, Project Development, Project Testing, Project Documentation and Project Demonstration.
- Team requirement: minimum 2 members, standard 4 members, maximum 5 members.
- Future direction from the project concept: the application is designed as a scalable/adaptable AI-powered fitness solution that can evolve with user preferences and progress.